Project Summary: StockMarket App Deployment (Ansible, Terraform, Cloud Architecture)

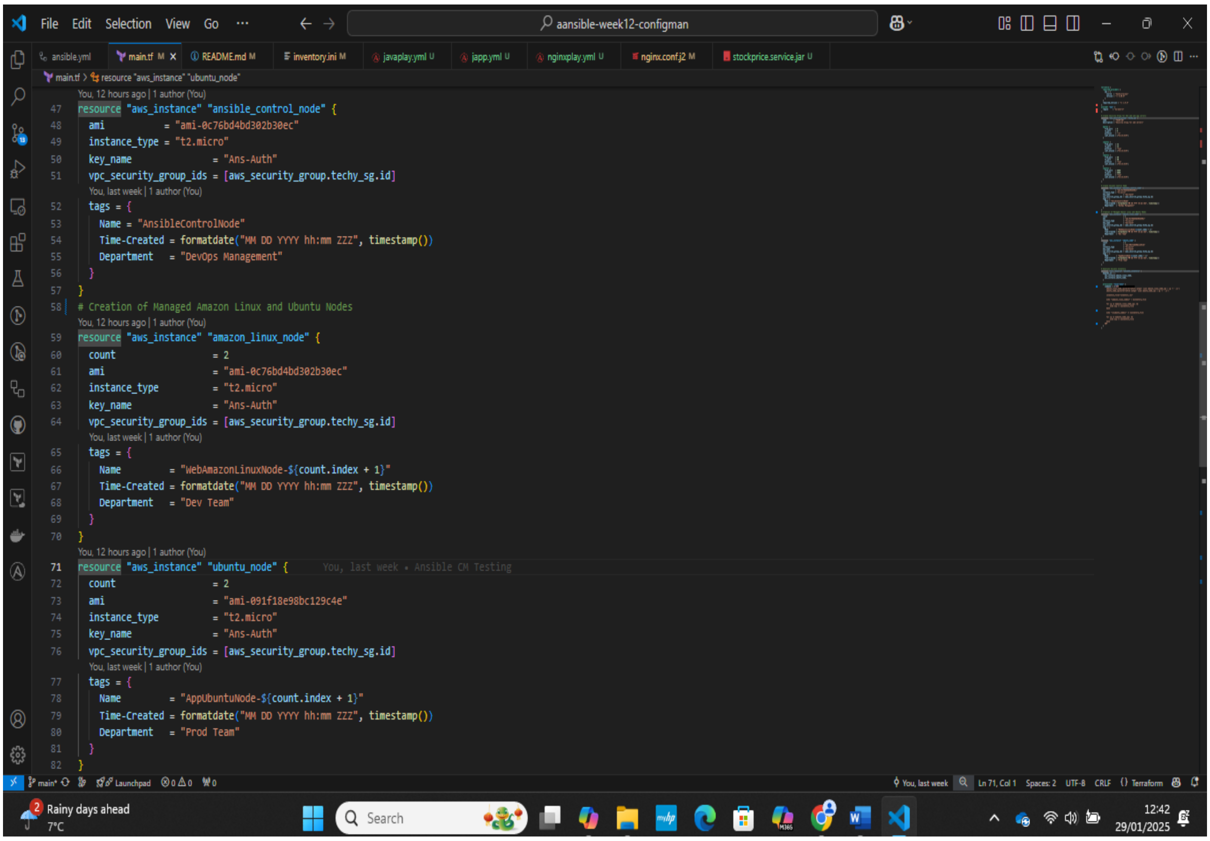
As the **DevOps Engineer** for this project, I successfully deployed the **StockMarket App**, which comprises **two Java backend systems** and **two Nginx reverse proxy servers**, all hosted within a **cloud infrastructure**. The project was structured into **three distinct levels**, focusing on **infrastructure provisioning, automation of configurations, and ensuring efficient traffic routing between Nginx and Java services**.

Step-by-Step Implementation

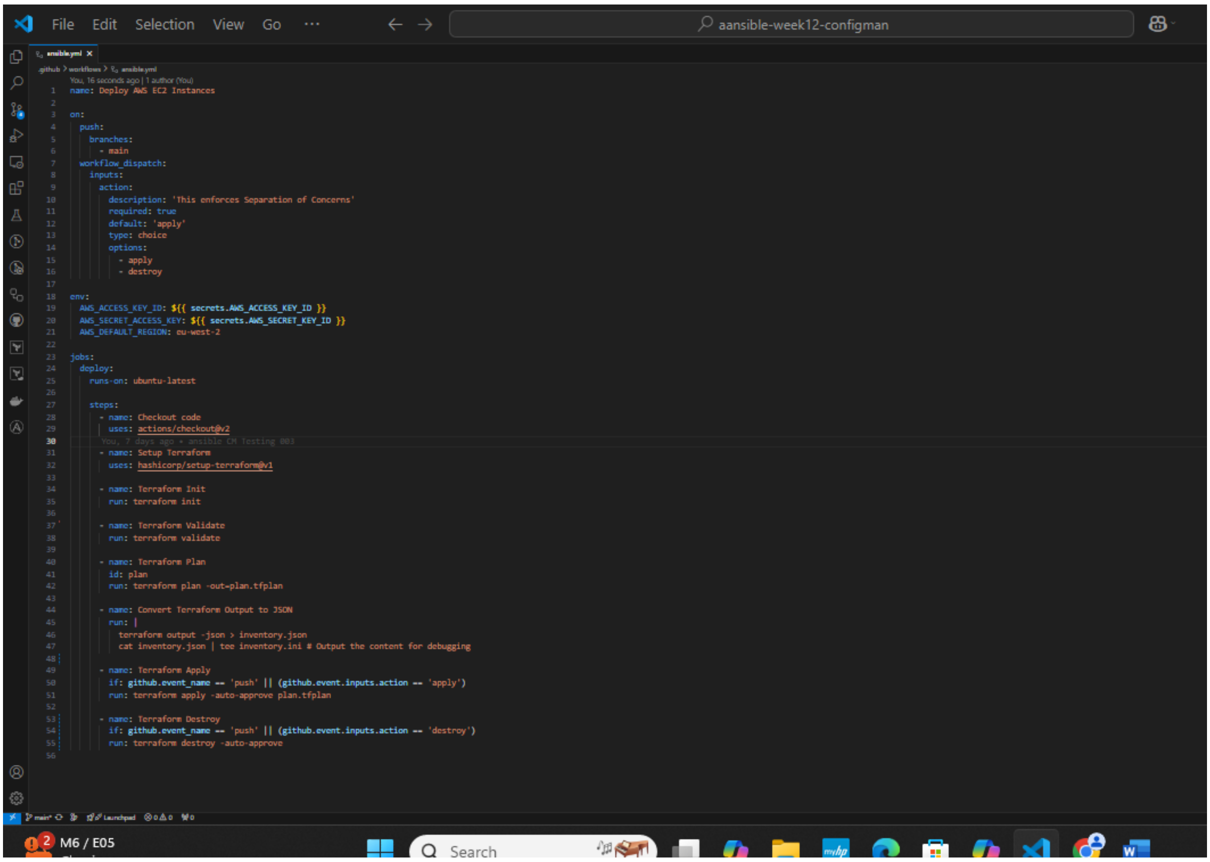
Level 1: Infrastructure Setup (Automated with Terraform)

✅ **Provision 5 Nodes (2 Nginx, 2 Java, 1 Database/Worker Node)** You can use **Terraform** to automate cloud instance creation or manually deploy them.

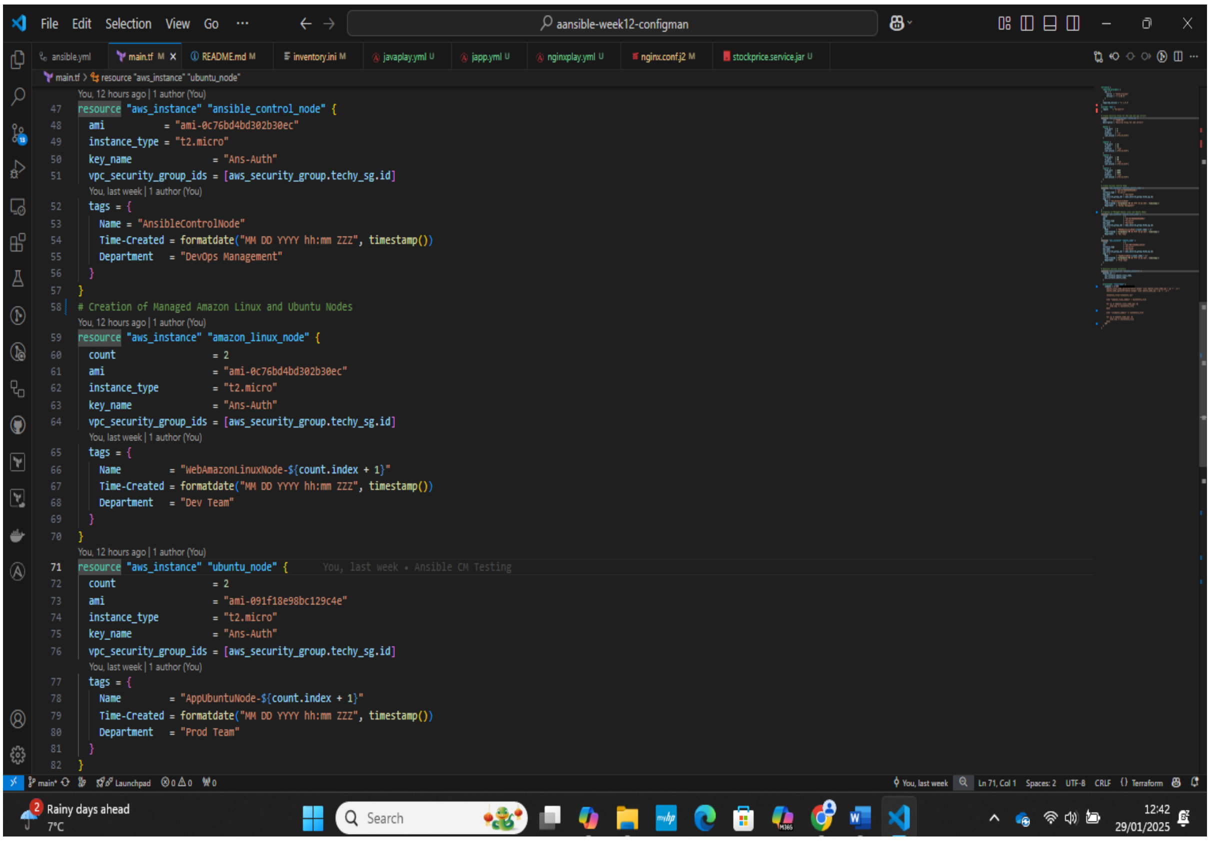
Terraform-Based Node Creation (main.tf)



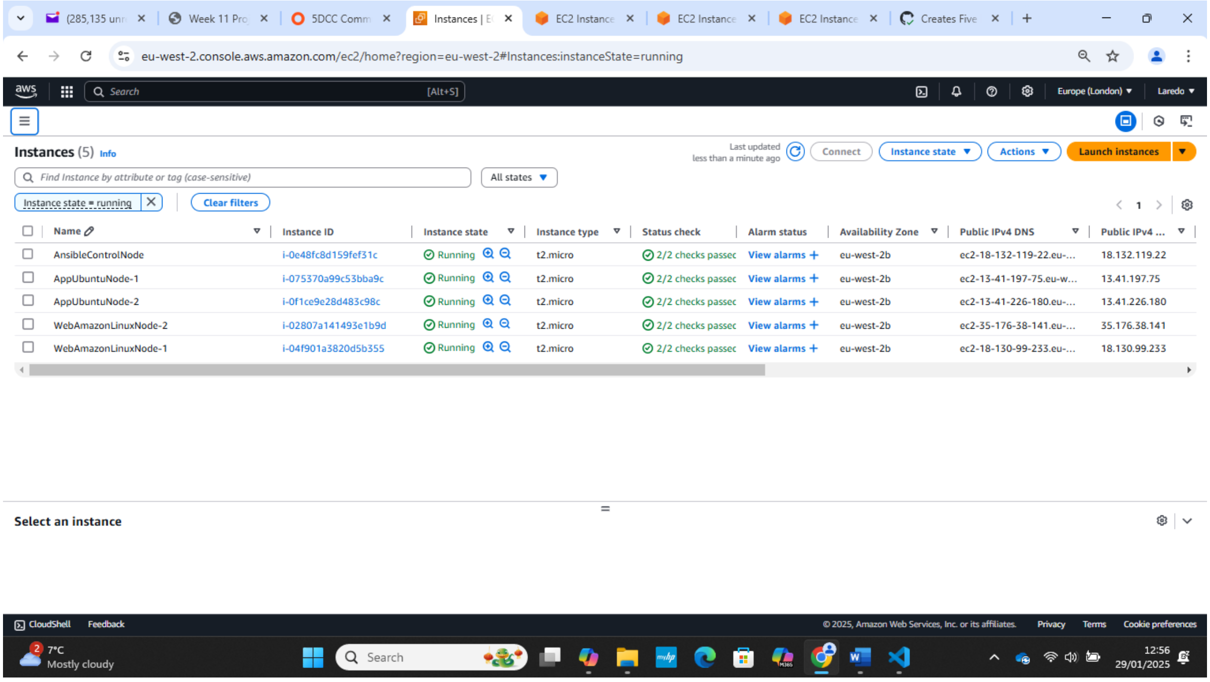
GitHub Actions Workflow



Workflow success



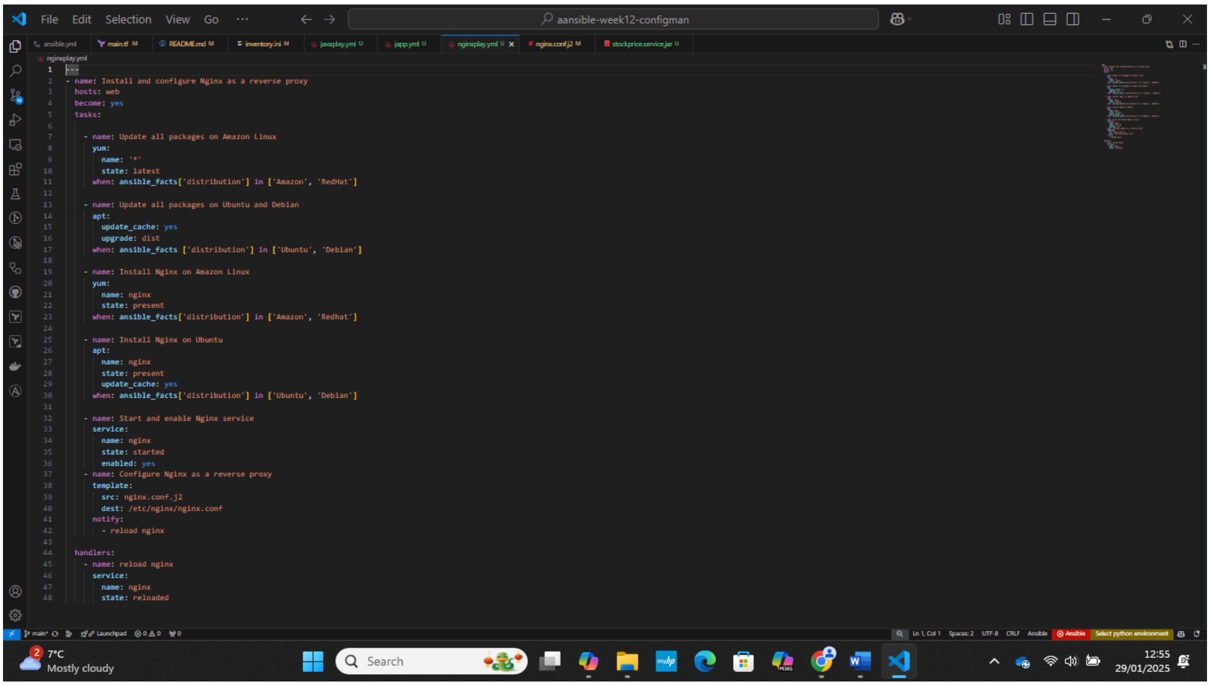
AWS Console showing Nodes Provisioned via Terraform



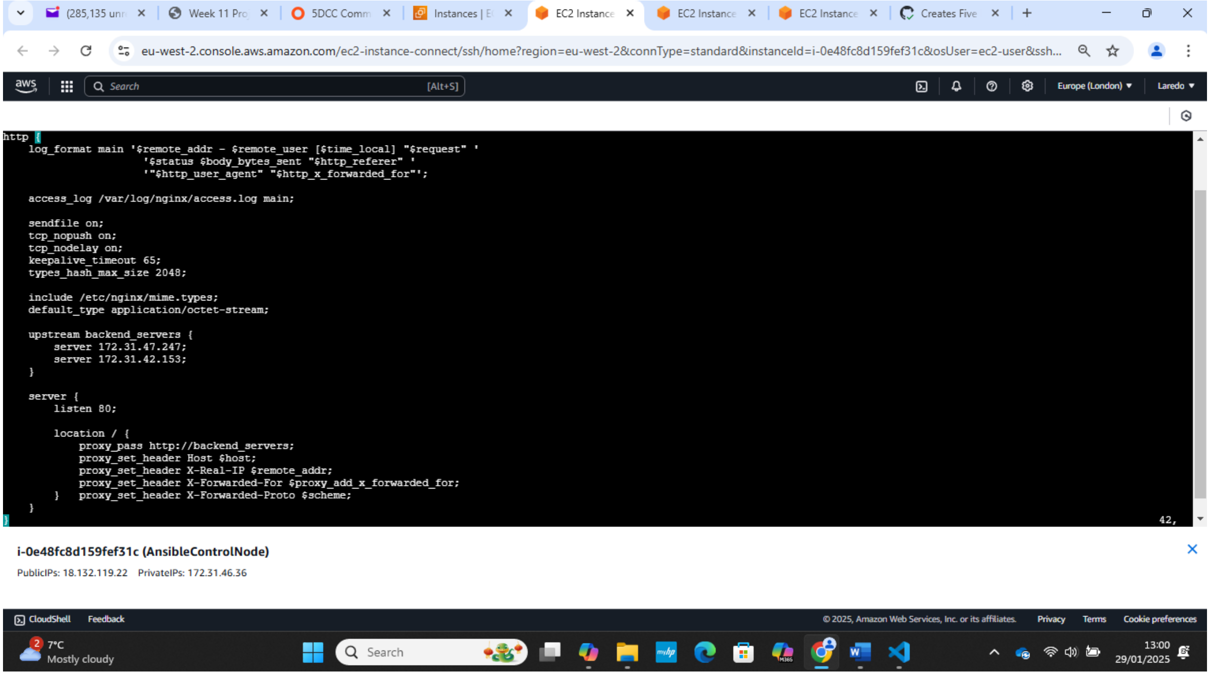
Level 2: Nginx Configuration with Ansible

✅ Install and Configure Nginx

Playbook to install Nginx (and Configure as REVERSE PROXY) on Nodes

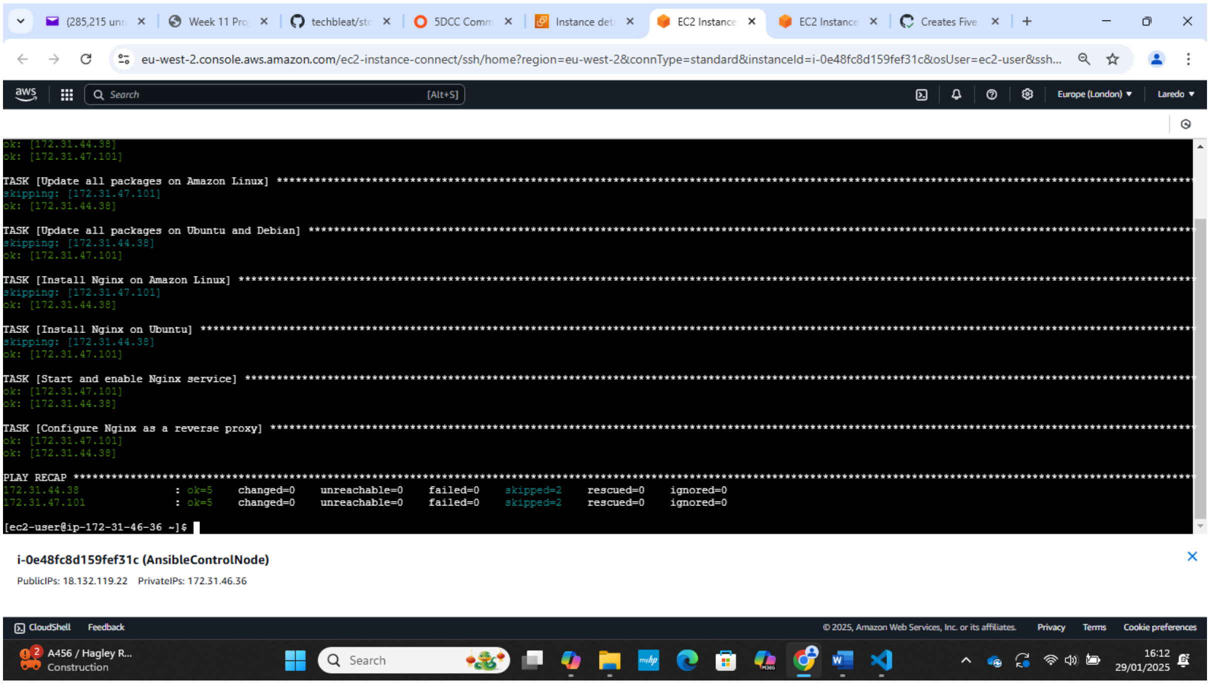


✅ Nginx Reverse Proxy Template (nginx.conf.j2)

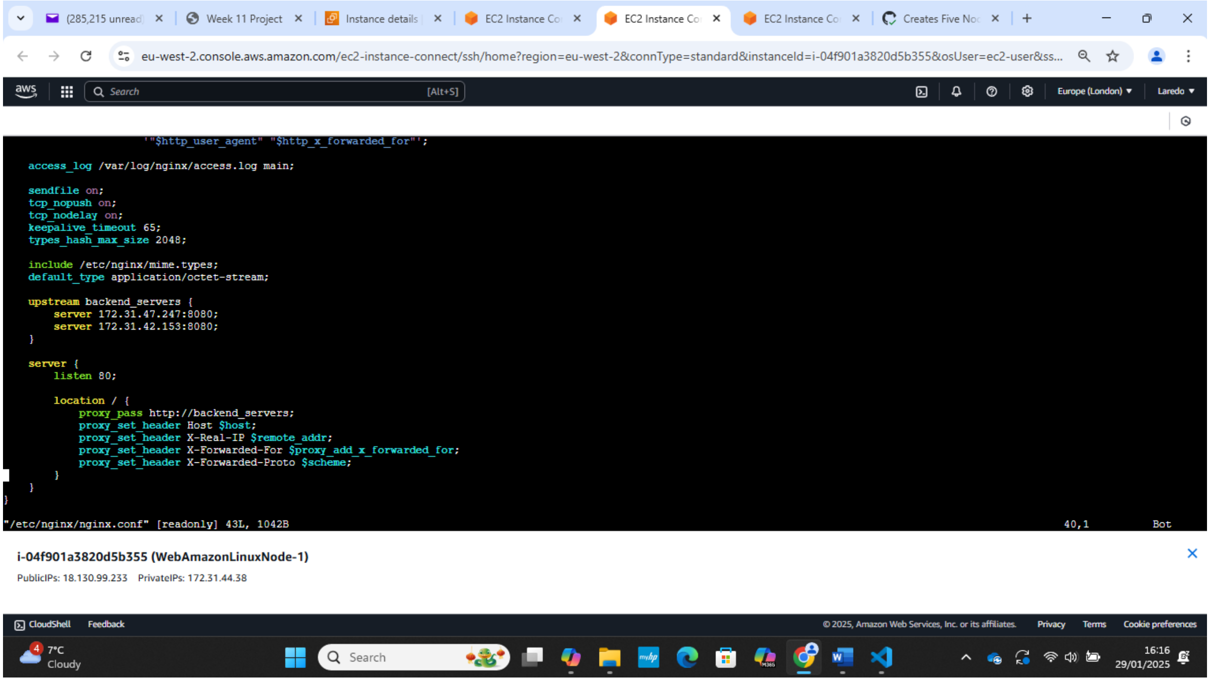


✅ Run: ansible-playbook

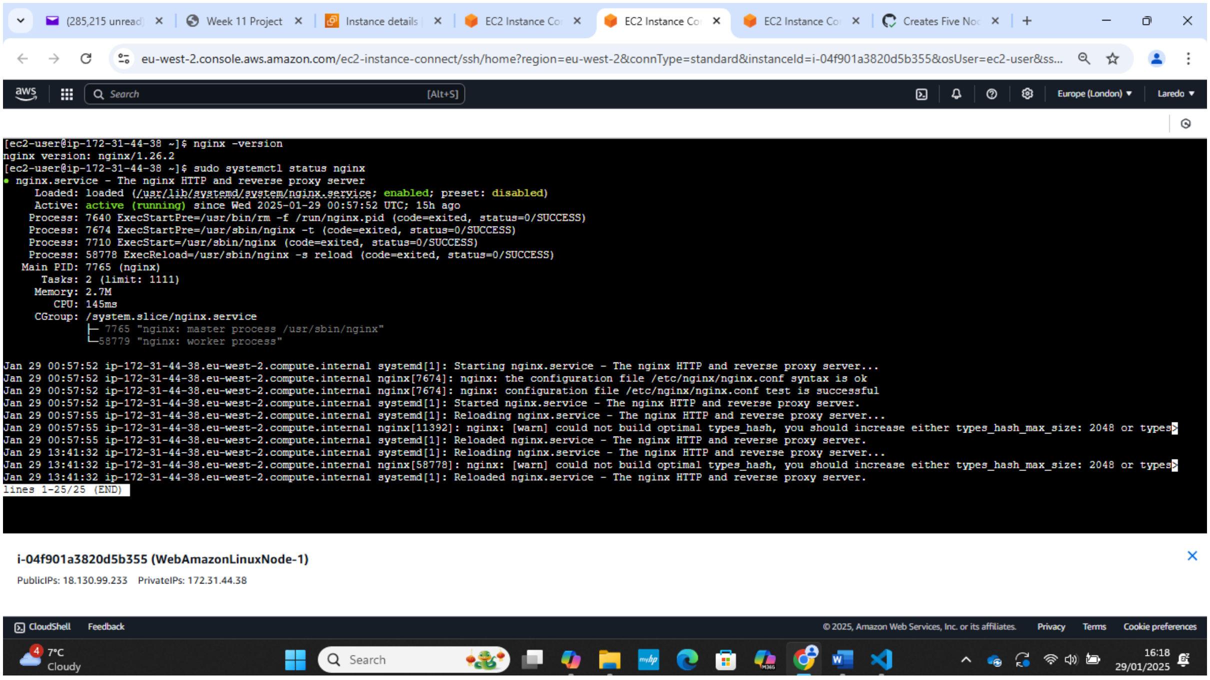
Nginx service setup as a REVERSE PROXY for the Java service: Success



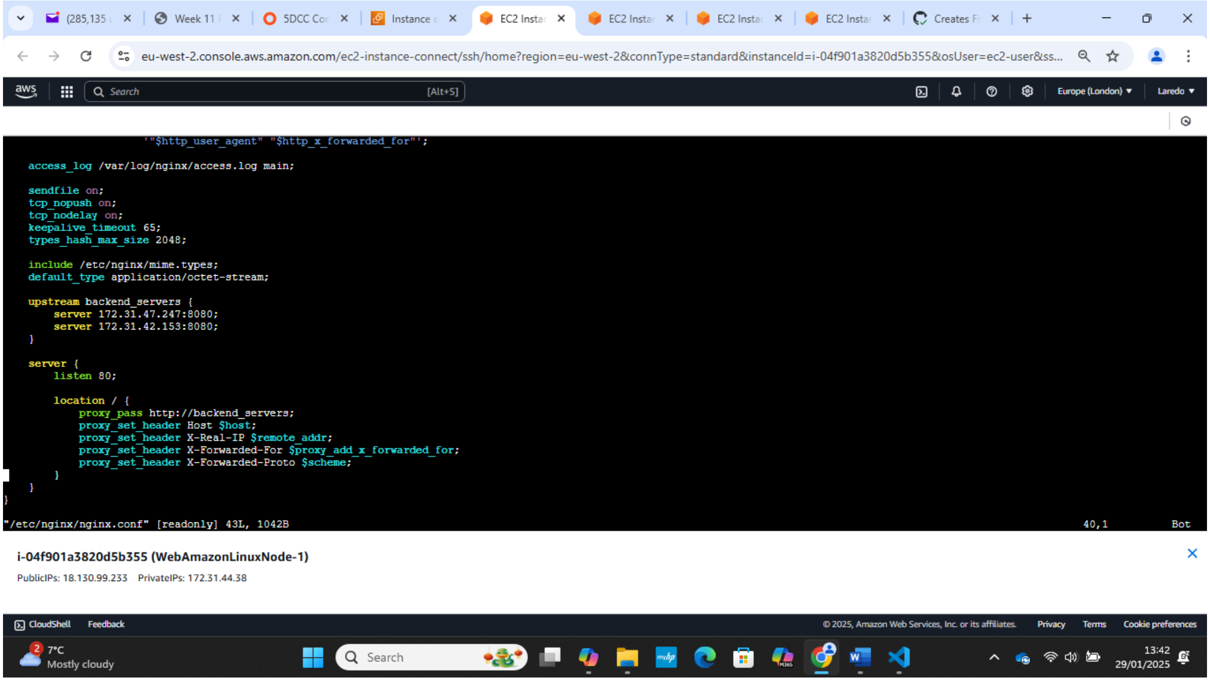
Reverse proxy for WebAmazonLinuxNode-1 (also AppUbuntuNode-1)



Nginx service running on WebAmazonLinuxNode-1 (also AppUbuntuNode-1)



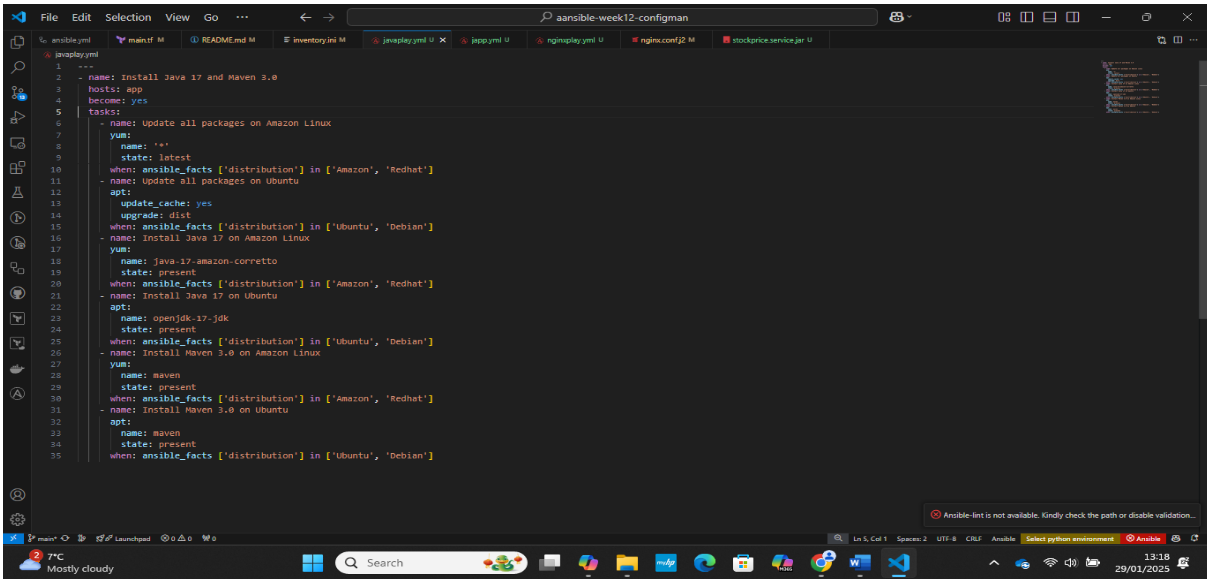
Nginx Reverse proxies forward traffic to the two Java backend nodes on port 8080



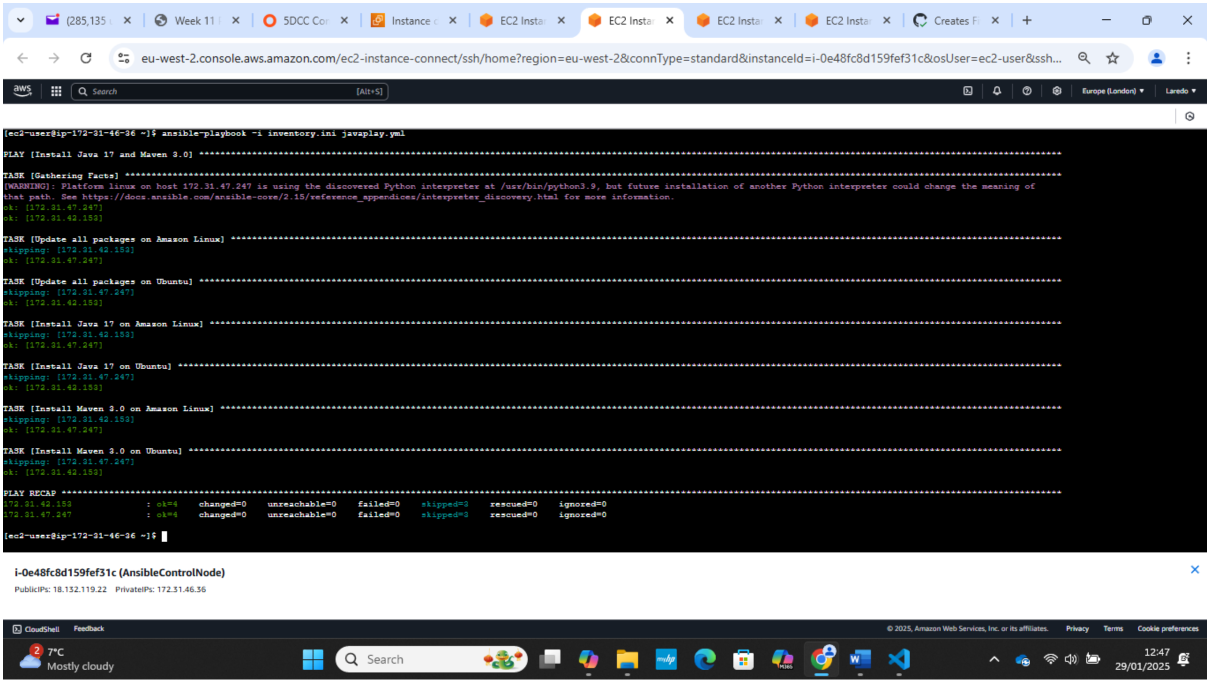
Level 3: Java Backend Deployment with Ansible

✅ Install Java & Maven

Playbook to install and configure Java 17 and Maven 3.0 on Backend Java nodes.

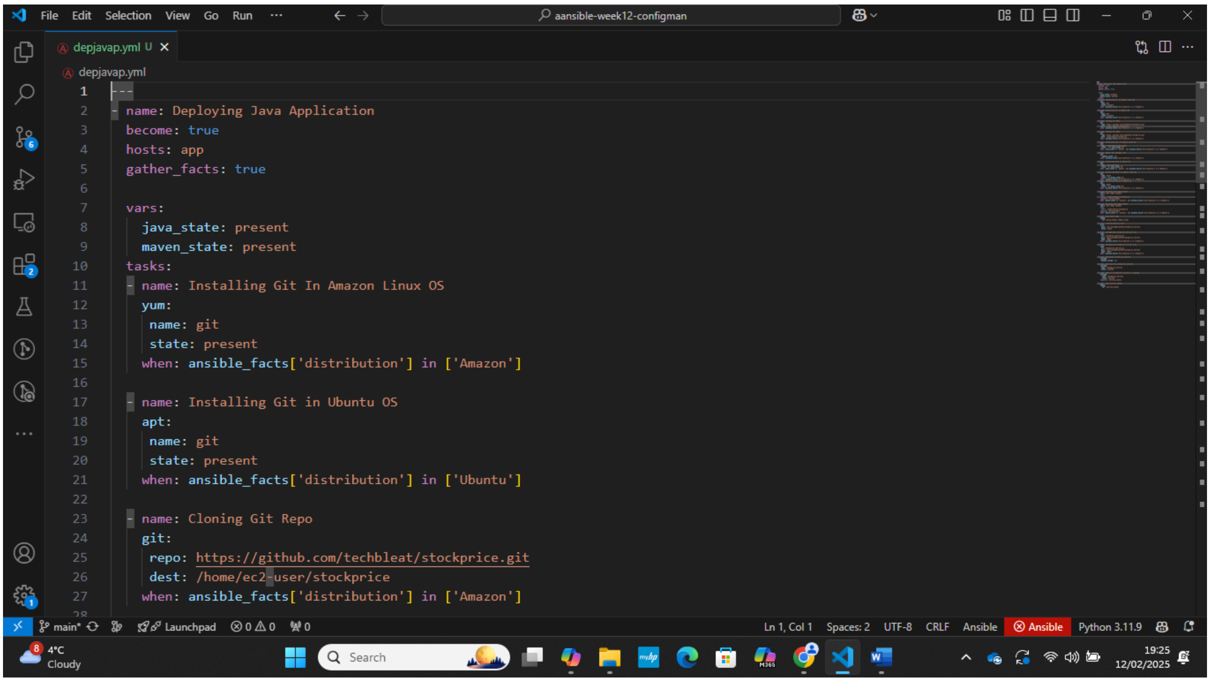


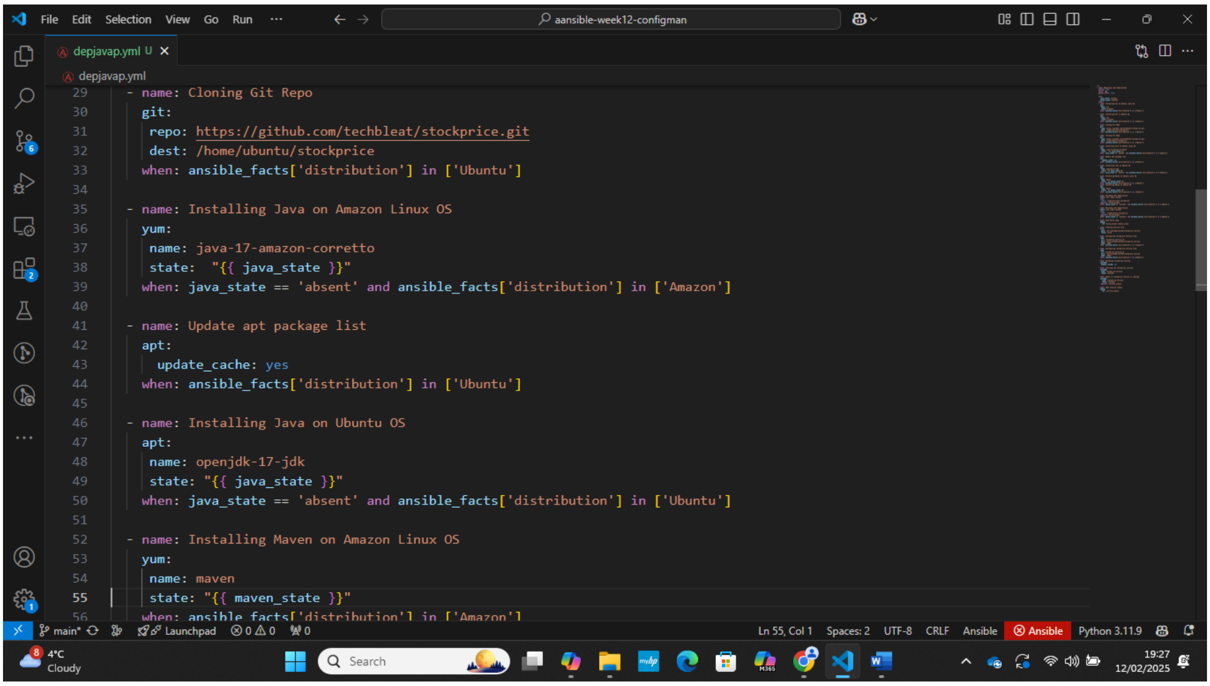
Java & Maven Installed on Amazon and Ubuntu backend App nodes.

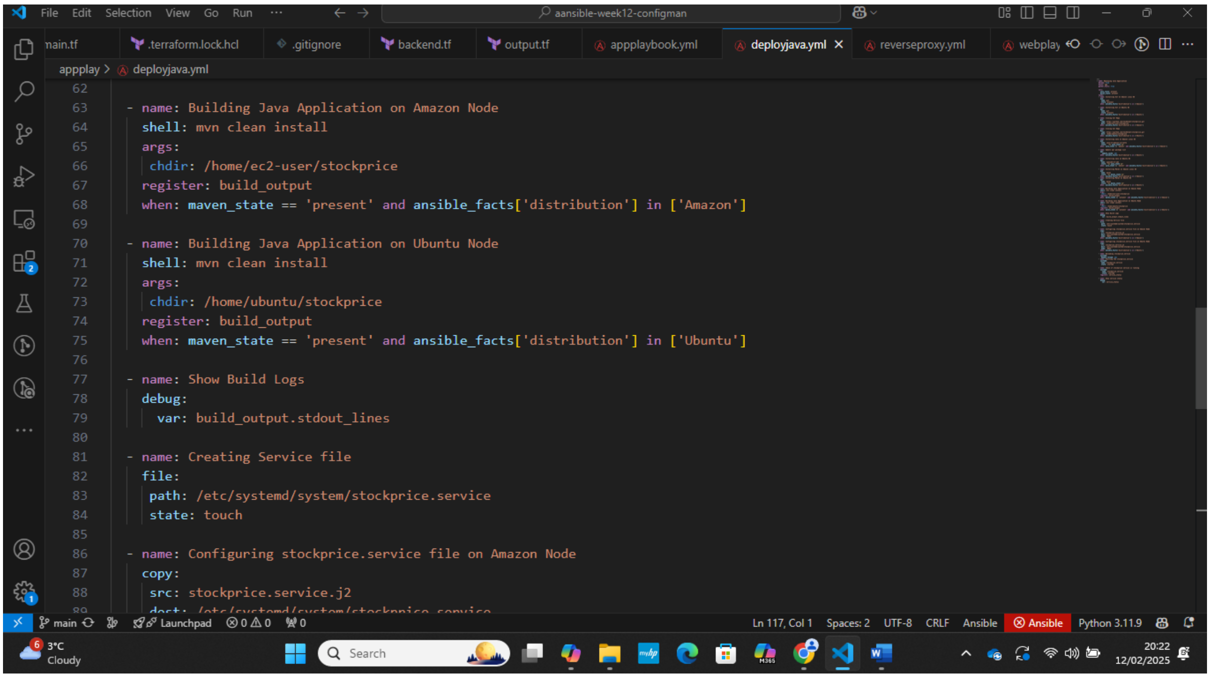


✅ Deploy Java Application

Playbook to deploy the Java app on the Java Backend nodes







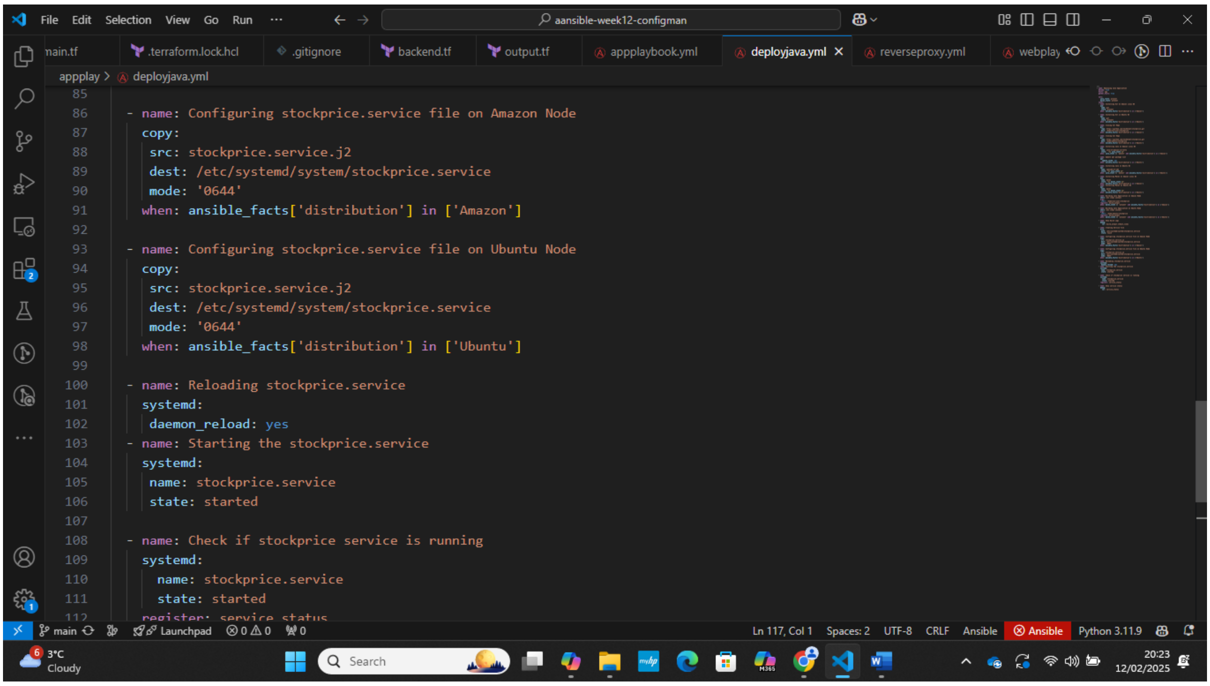
- name: Building Java Application on Amazon Node
  shell: mvn clean install
  args:
    chdir: /home/ec2-user/stockprice
  register: build_output
  when: maven_state == 'present' and ansible_facts['distribution'] in ['Amazon']

- name: Building Java Application on Ubuntu Node
  shell: mvn clean install
  args:
    chdir: /home/ubuntu/stockprice
  register: build_output
  when: maven_state == 'present' and ansible_facts['distribution'] in ['Ubuntu']

- name: Show Build Logs
  debug:
    var: build_output.stdout_lines

- name: Creating Service file
  file:
    path: /etc/systemd/system/stockprice.service
    state: touch

- name: Configuring stockprice.service file on Amazon Node
  copy:
    src: stockprice.service.j2

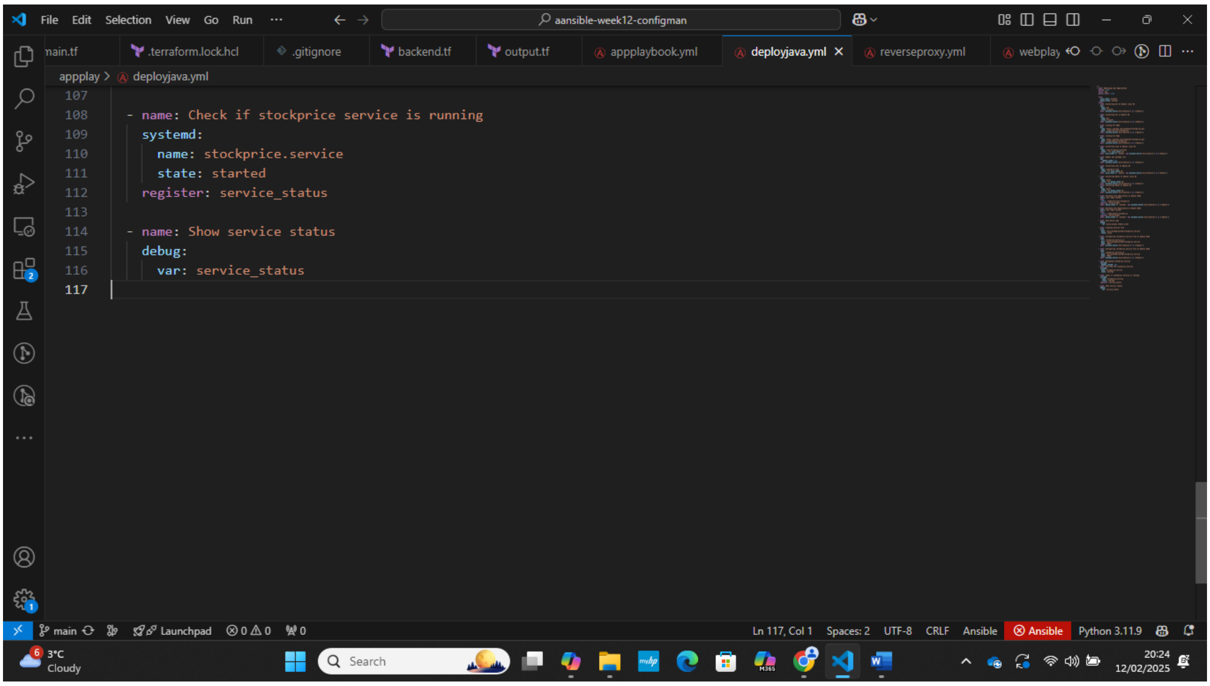


- name: Configuring stockprice.service file on Amazon Node
  copy:
    src: stockprice.service.j2
    dest: /etc/systemd/system/stockprice.service
    mode: '0644'
  when: ansible_facts['distribution'] in ['Amazon']

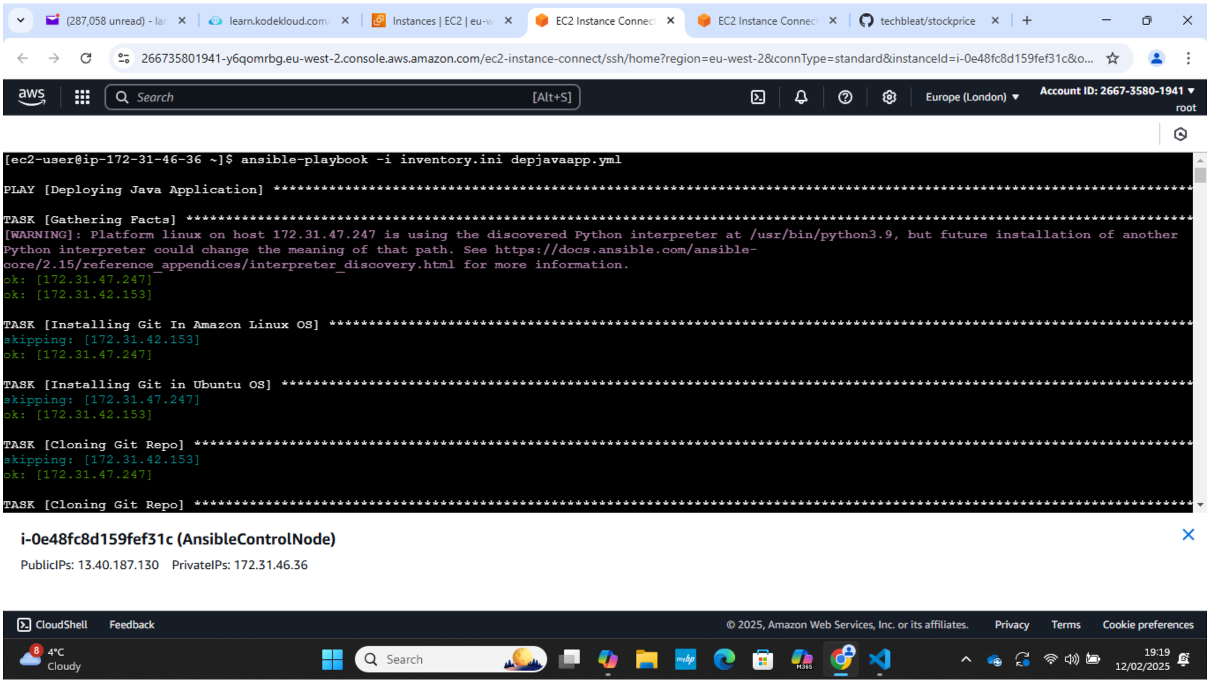
- name: Configuring stockprice.service file on Ubuntu Node
  copy:
    src: stockprice.service.j2
    dest: /etc/systemd/system/stockprice.service
    mode: '0644'
  when: ansible_facts['distribution'] in ['Ubuntu']

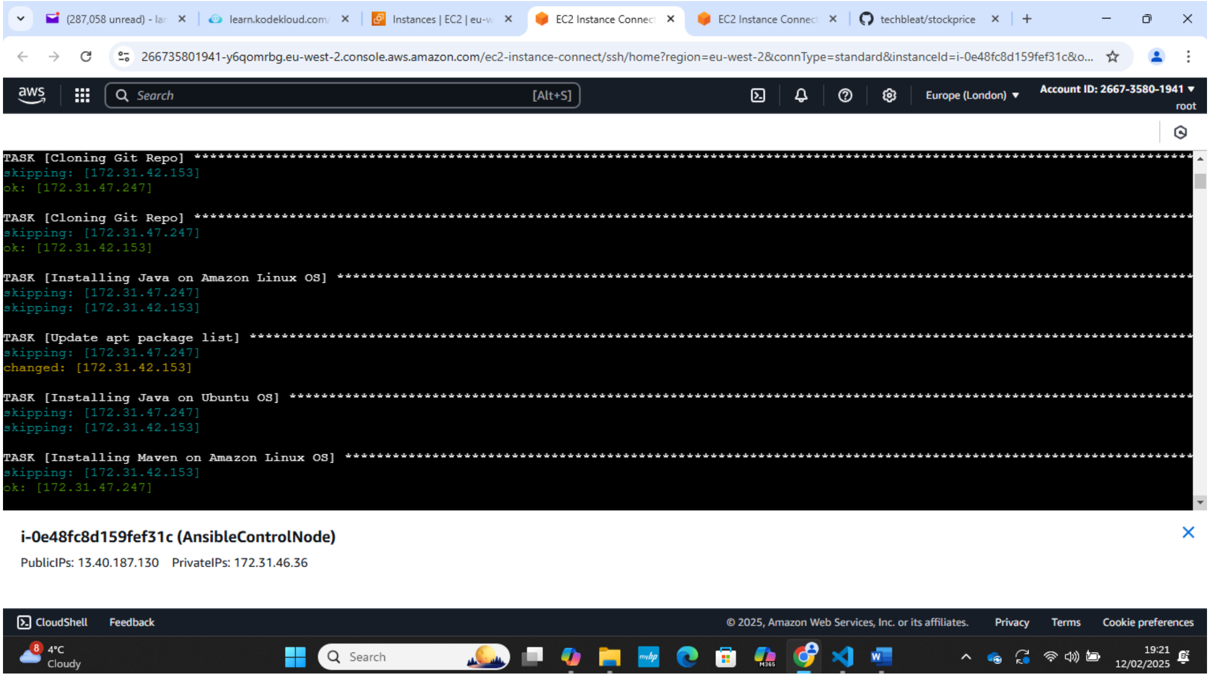
- name: Reloading stockprice.service
  systemd:
    daemon_reload: yes
- name: Starting the stockprice.service
  systemd:
    name: stockprice.service
    state: started

- name: Check if stockprice service is running
  systemd:
    name: stockprice.service
    state: started



✅ Run: ansible-playbook for Deployment





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root
TASK [Cloning Git Repo]
skipping: [172.31.42.153]
ok: [172.31.47.247]
TASK [Cloning Git Repo]
skipping: [172.31.47.247]
ok: [172.31.42.153]
TASK [Installing Java on Amazon Linux OS]
skipping: [172.31.47.247]
skipping: [172.31.42.153]
TASK [Update apt package list]
skipping: [172.31.47.247]
changed: [172.31.42.153]
TASK [Installing Java on Ubuntu OS]
skipping: [172.31.47.247]
skipping: [172.31.42.153]
TASK [Installing Maven on Amazon Linux OS]
skipping: [172.31.42.153]
ok: [172.31.47.247]
i-0e48fc8d159fef31c (AnsibleControlNode)
PublicIPs: 13.40.187.130 PrivateIPs: 172.31.46.36
CloudShell
Feedback
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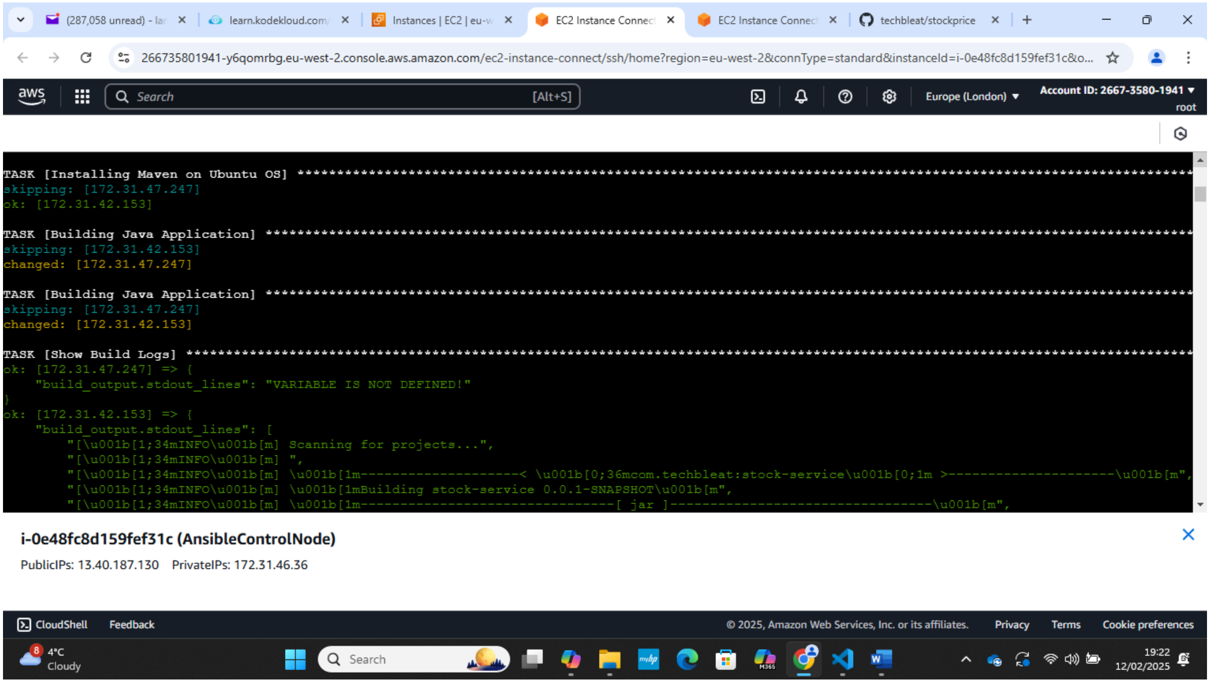

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Europe (London)
Account ID: 2667-3580-1941
root
TASK [Installing Maven on Ubuntu OS]
skipping: [172.31.47.247]
ok: [172.31.42.153]
TASK [Building Java Application]
skipping: [172.31.42.153]
changed: [172.31.47.247]
TASK [Building Java Application]
skipping: [172.31.47.247]
changed: [172.31.42.153]
TASK [Show Build Logs]
ok: [172.31.47.247] => {
"build_output.stdout_lines": "VARIABLE IS NOT DEFINED!"
}
ok: [172.31.42.153] => {
"build_output.stdout_lines": [
"[\u001b[1;34mINFO\u001b[m] Scanning for projects...",
"[\u001b[1;34mINFO\u001b[m] ",
"[\u001b[1;34mINFO\u001b[m] \u001b[1m--------------------< \u001b[0;36mcom.techbleat:stock-service\u001b[0;1m >--------------------\u001b[m",
"[\u001b[1;34mINFO\u001b[m] \u001b[1mBuilding stock-service 0.0.1-SNAPSHOT\u001b[m",
"[\u001b[1;34mINFO\u001b[m] \u001b[1m--------------------------------[jar]---------------------------------\u001b[m",
i-0e48fc8d159fef31c (AnsibleControlNode)
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CloudShell
Feedback
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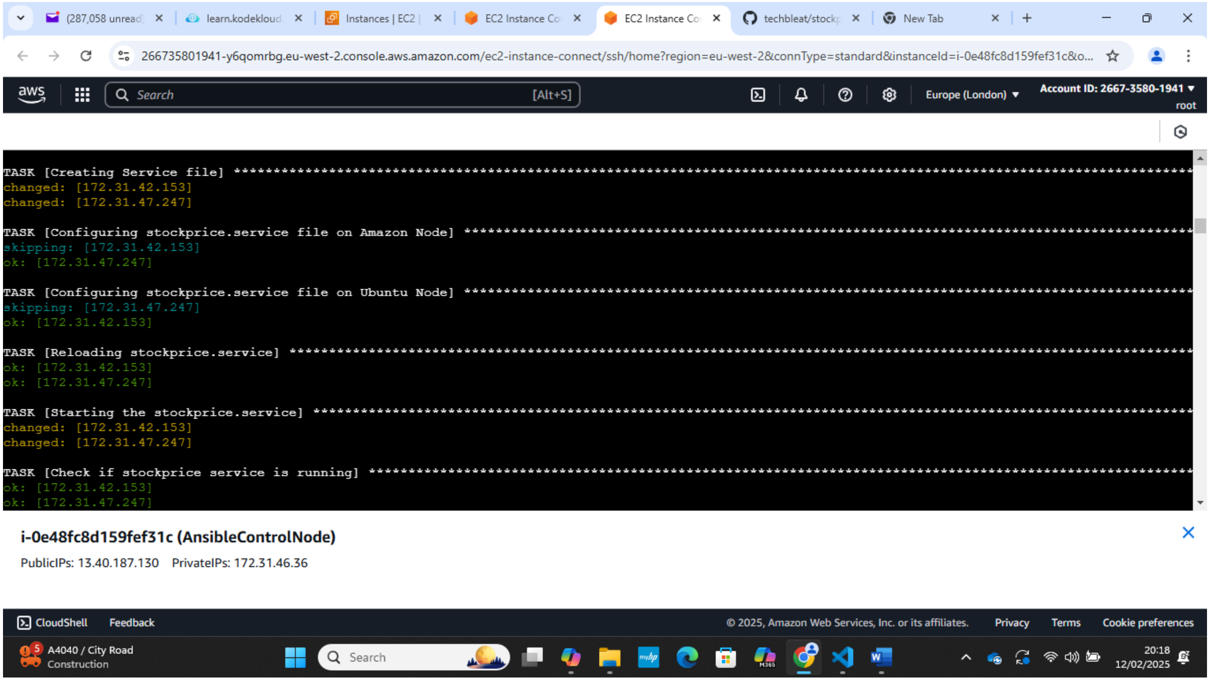
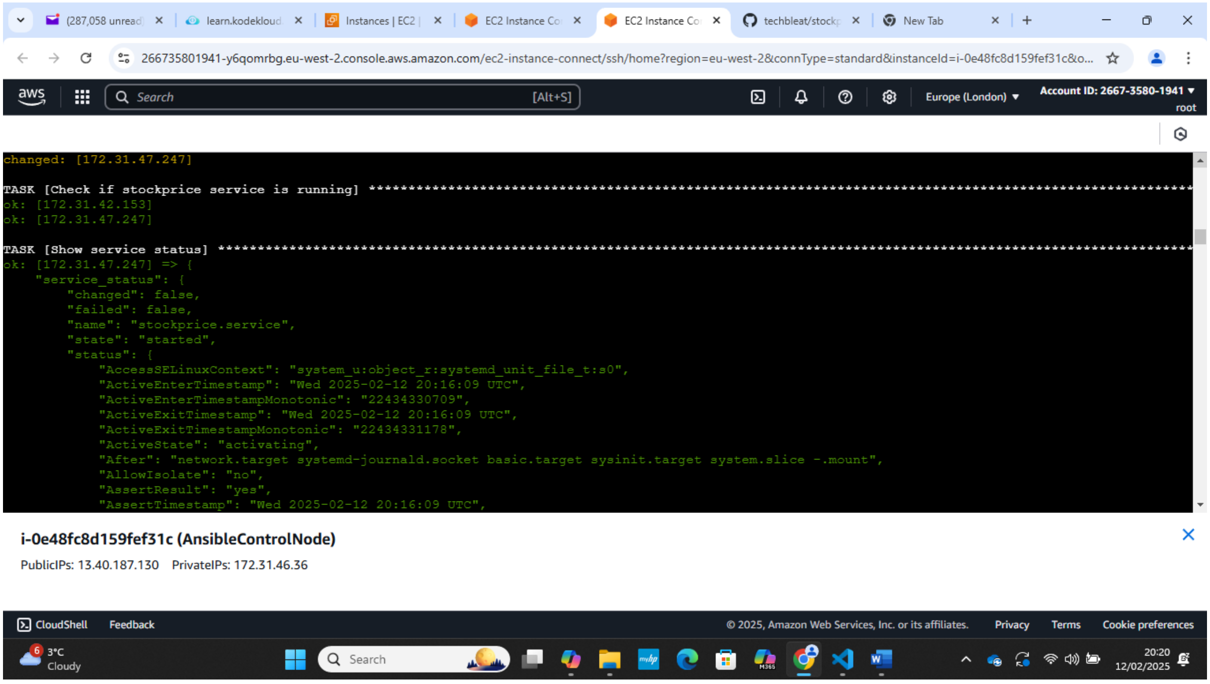
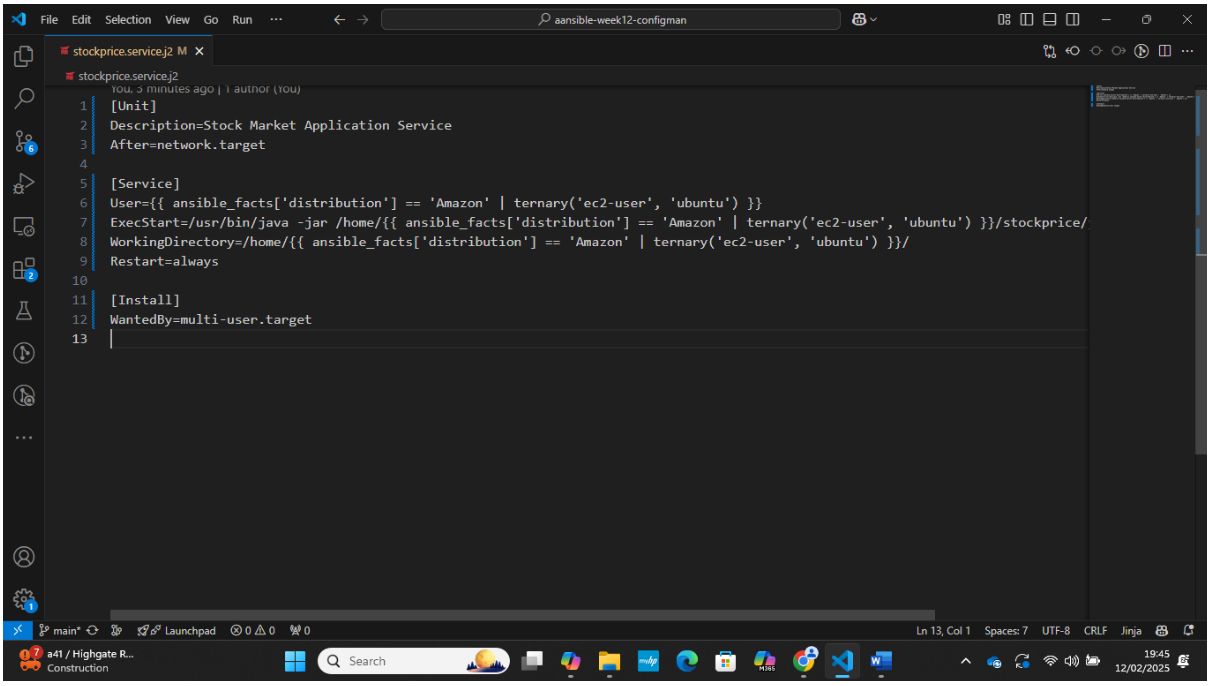

TASK [Creating Service file]
changed: [172.31.42.153]
changed: [172.31.47.247]
TASK [Configuring stockprice.service file on Amazon Node]
skipping: [172.31.42.153]
ok: [172.31.47.247]
TASK [Configuring stockprice.service file on Ubuntu Node]
skipping: [172.31.47.247]
ok: [172.31.42.153]
TASK [Reloading stockprice.service]
ok: [172.31.42.153]
ok: [172.31.47.247]
TASK [Starting the stockprice.service]
changed: [172.31.42.153]
changed: [172.31.47.247]
TASK [Check if stockprice service is running]
ok: [172.31.42.153]
ok: [172.31.47.247]
i-0e48fc8d159fef31c (AnsibleControlNode)
PublicIPs: 13.40.187.130 PrivateIPs: 172.31.46.36


changed: [172.31.47.247]
TASK [Check if stockprice service is running]
ok: [172.31.42.153]
ok: [172.31.47.247]
TASK [Show service status]
ok: [172.31.47.247] => {
"service_status": {
"changed": false,
"failed": false,
"name": "stockprice.service",
"state": "started",
"status": {
"AccessSELinuxContext": "system_u:object_r:systemd_unit_file_t:s0",
"ActiveEnterTimestamp": "Wed 2025-02-12 20:16:09 UTC",
"ActiveEnterTimestampMonotonic": "22434330709",
"ActiveExitTimestamp": "Wed 2025-02-12 20:16:09 UTC",
"ActiveExitTimestampMonotonic": "22434331178",
"ActiveState": "activating",
"After": "network.target systemd-journald.socket basic.target sysinit.target system.slice -.mount",
"AllowIsolate": "no",
"AssertResult": "yes",
"AssertTimestamp": "Wed 2025-02-12 20:16:09 UTC",
i-0e48fc8d159fef31c (AnsibleControlNode)
PublicIPs: 13.40.187.130 PrivateIPs: 172.31.46.36

Created a service wrapper around the java application to easily manage it



Level 4: Verification of Traffic Routing

✅ **Test Nginx Proxy Forwarding**

curl -v http://<NGINX_IP>:configured_port

✅ **Check Java Logs**

java -jar target/stockprice.jar

Next Steps

- ✅ Full automated **CI/CD deployment**
- ✅ **Integration of Cloud Load Balancer for scalability**
- ✅ **Implementation of monitoring with Prometheus/Grafana**